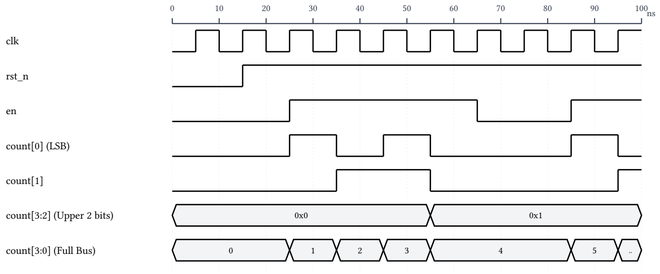
Signal Slicing, Interpreters & Warning System

This example demonstrates the advanced signal slicing, data interpreters (including 7-segment display auto-decoding), and warning indicators in **signel**.

1. Bit Slicing: Single Bits and Sub-Bus Ranges

In standard VCD files, signals like `count [3:0]` are declared as multi-bit vectors. **signel** allows you to slice individual bits (`count[0]`, `count[1]`, etc.) to view them as 1-bit digital waveforms, or extract sub-bus slices (`count[3:2]`) as buses without modifying the simulation source.

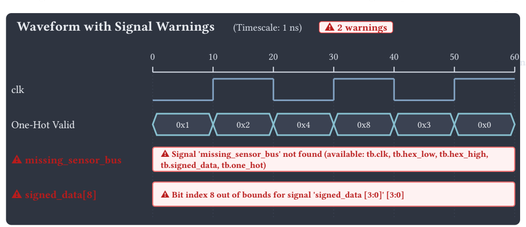
Synchronous Counter — Bit Slicing Demo (Timescale: 1 ns)



2. 7-Segment Display Auto-Decoding & Interpreters

When designing FPGA or digital circuits driving 7-segment displays (such as on Altera/Intel DE-series or Xilinx boards), signals represent segment activations (a, b, c, d, e, f, g). **signel** includes dedicated interpreters that decode active-low or active-high 7-segment patterns into readable hex characters (`0`–`F`). If an invalid pattern is encountered, **signel** highlights the error in the waveform and logs a warning!

Interpreters Demo: 7-Segment, Signed 2's Complement & One-Hot (Timescale: 1 ns) ⚠ 3 warnings

3. Robust Error Reporting & Missing Signal Warnings

If a signal cannot be found or if an index is out of bounds, **signel** produces an in-place visual warning banner with the list of available signals, without crashing your document build. You can also enforce strict verification with `strict: true`.